

GAURISH BANGERA

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PROFESSIONAL SUMMARY

Computer Science & Engineering undergraduate at NMAMIT with strong expertise in Data Structures, Algorithms, Full-Stack Web Development, and AI Engineering. Hands-on experience developing Local LLM privacy analyzers, custom database engines in C++, full-stack Go/React web platforms, and deep learning vision systems. Certified in Microsoft Azure AI Fundamentals with proven research and industry internship experience.

EDUCATION

NMAM Institute of Technology, Nitte, Karkala

Expected May 2027

Bachelor of Technology (B.Tech) in Computer Science & Engineering

TECHNICAL SKILLS

Programming Languages: C++, Python, Go, Java, C, JavaScript, TypeScript, SQL

AI & Machine Learning: PyTorch, Deep Learning, Local LLMs (Ollama / Transformers), Computer Vision, Scikit-Learn, Pandas

Web Development: React, Node.js, HTML5, CSS3, RESTful APIs, TypeScript

Databases & Tools: MySQL, PostgreSQL, XAMPP, Git, GitHub, VS Code

Core Competencies: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Systems, Compiler Design

INTERNSHIP EXPERIENCE

NITK (National Institute of Technology Karnataka) — Research Intern — RGB-T Vision Portal

2025

- Designed and deployed the **RGB-T Vision Portal** ([Live Portal](#)), an interactive web platform presenting thermal-visible computer vision research methodologies and datasets.
- Integrated PyTorch research documentation, interactive simulations, and multi-modal vision datasets to streamline project outcomes for academic collaborators.
- Engineered responsive UI components utilizing modern web standards to visualize complex thermal-RGB fusion experimental pipelines.

NativeSoftTech — Web Development Intern

June 2025 – July 2025

- Developed dynamic, responsive web interfaces and reusable user components adhering to modern HTML5, CSS3, and JavaScript standards.
- Collaborated on client-facing frontend features, optimizing page layout responsiveness, cross-browser compatibility, and overall site performance.
- Refactored frontend code modules to enhance software maintainability and user engagement metrics.

CodSoft — Data Science Virtual Intern

June 2025 – July 2025

- Conducted data preprocessing, exploratory data analysis (EDA), and feature engineering on structured datasets.
- Implemented predictive machine learning models to analyze data patterns and evaluate algorithm performance using statistical metrics.

PROJECTS

Consent-Lens (Offline Permission & Consent Impact Analyzer)

Python, Local LLMs, Ollama, GenAI

- Architected an offline privacy and permission impact analyzer using local LLMs to evaluate app permissions and privacy terms without transmitting user data online.
- Engineered structured prompt pipelines and risk scoring algorithms for automated permission policy analysis and mitigation reporting.

College Database System Engine

C++, Custom Data Structures, File I/O

- Engineered a high-performance in-memory database engine in C++ utilizing custom data structures for student, faculty, and fee record management without external STL database wrappers.
- Implemented binary file persistence, custom search indexing, and automated report generation modules to optimize data lookup and storage efficiency.

Intelligent Contact Management System

Go, React, TypeScript, PostgreSQL

- Architected a full-stack contact management application using **Go** for high-concurrency backend services and **React + TypeScript** for a responsive interface.

- Designed relational database schemas and optimized **PostgreSQL** queries for fast CRUD operations, full-text search, and soft-delete/restore functionality.
- Engineered REST APIs following Clean Architecture principles to guarantee code maintainability and system scalability.

Compiler for Hypothetical Language

Python, Compiler Design, LR Parsing

- Built an SLR(1) parser and execution engine for a custom mini programming language, executing lexical analysis, parsing, and syntax evaluation.
- Computed FIRST/FOLLOW sets and constructed LR(0) state machines, resolving parsing table conflicts for deterministic syntax evaluation.

Smart Insurance Cost Prediction

Python, Machine Learning, Scikit-Learn

- Developed an end-to-end machine learning pipeline using Kaggle insurance datasets to predict health insurance premiums based on user attributes.
- Benchmarked Random Forest, KNN, and Linear Regression models using regression metrics (R^2 , RMSE) to select the optimal model.

ACHIEVEMENTS & CERTIFICATIONS

- **Microsoft Certified:** Azure AI Fundamentals (AI-900).
- **Technical Learning:** Completed C++ Professional Certification from Infosys Springboard; attended Microsoft Azure Studio webinars.
- **Competitive Programming:** Active problem solver on LeetCode and TakeUForward (TUF) focusing on Data Structures & Algorithms.
- **Leadership & Community:** Awarded Certificate of Appreciation by Tudar Club for organizing and executing the Tululipi event.